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by

AMRUTH M KASHYAP	**USN:1BI23ET004**
CHIRANTH B M	**USN:1BI23ET012**
NIHAL S JAMAKHANDI	**USN:1BI23ET039**
PRAJWAL S	**USN:1BI23ET043**

Under the guidance of

THYAGARAJ R

Assistant Professor

Dept. of Electronics and Telecommunication Engineering

B.I.T., Bengaluru

DEPARTMENT OF ELECTRONICS & TELECOMMUNICATION ENGINEERING

BANGALORE INSTITUTE OF TECHNOLOGY, BENGALURU – 560004

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BANGALORE INSTITUTE OF TECHNOLOGY

K R Road, V V Pura, Bengaluru-560004

Department of Electronics and Telecommunication Engineering

CERTIFICATE

Certified that the Project work Phase-1 entitled **Real Time Vision-Based Precision Landing of UAV in GPS-Limited Environments** work carried out by **Amruth M Kashyap (1BI23ET004), Chiranth B M (1BI23ET012), Nihal S Jamakhandi (1BI23ET039), Prajwal S (1BI23ET043)** bona fide students of **Bangalore Institute of Technology** in partial fulfillment for the award of **Bachelor of Engineering** Degree in Electronics and Telecommunication Engineering of the Visvesvaraya Technological University, Belagavi during the year 2025-26. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the Departmental Library. The Project Report has been approved as it satisfies the academic requirements in respect of Project Work prescribed for the said Degree.

THYAGARAJ R
Project Guide

Dr. Rajeswari M
Professor & HOD

Dr.Vijay Prakash A M
Principal, BIT

Final Assessment

Name of the Evaluators	Signature with Date
Evaluator 1	
Evaluator 2	

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ABSTRACT

The growing deployment of Unmanned Aerial Vehicles (UAVs) in time-critical applications such as infrastructure inspection, emergency response, and precision logistics demands robust autonomous landing capability independent of Global Navigation Satellite System (GNSS) availability. Conventional GPS-guided return-to-home procedures exhibit positional deviations of one to three meters under signal degradation, rendering them unsuitable for confined or obstacle-rich landing zones. This project presents the design, implementation, and experimental evaluation of a real-time vision-based precision landing system for a quad copter UAV operating in GPS-limited environments. The proposed architecture integrates the open-source Pixhawk 2.4.8 flight controller with an OpenMV H7 machine-vision camera that autonomously detects and tracks a nested AprilTag fiduciary marker affixed to the landing pad. Angular offset data computed onboard the camera is transmitted as MAVLink LANDING TARGET messages to the flight controller, which executes a closed-loop descent using its built-in Extended Kalman Filter and PID stabilization loops. Experimental results demonstrate a mean landing deviation of approximately 9 to 12 cm across outdoor, semi-outdoor, and indoor environments at descent altitudes of two and three meters, confirming the feasibility of cost-effective, GPS-independent precision landing at the sub-decimeter level.

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LIST OF ABBREVIATIONS

Abbreviation	**Description**
2-D	Two Dimensional
CAT	Computed Axial Tomography
EIS	Electrochemical Impedance Spectroscopy
fMRI	functional Magnetic Resonance Imaging
LED	Light Emitting Diode
MEA	Micro Electrode Array
MRI	Magnetic Resonance Imaging
PDMS	Polydimethylsiloxane
PET	Positron Emission Tomography
WHO	World Health Organization

Chapter 1

INTRODUCTION

1.1 Background

The operational envelope of modern unmanned aerial vehicles has expanded dramatically beyond their original military origins. Today, quadcopters are routinely employed in commercial delivery networks, precision agriculture, structural health monitoring, search-and-rescue coordination, and cinematic aerial photography. As autonomous flight missions grow in complexity, the terminal phase descent and touchdown emerges as the single most failure-prone segment of any sortie. Rough or imprecise landings account for a disproportionate share of airframe damage, mission aborts, and payload loss events recorded across consumer and professional UAV fleets. Conventional landing guidance relies on Global Navigation Satellite System (GNSS) receivers to maintain positional awareness during descent. Under ideal open-sky conditions, a consumer-grade GPS module provides horizontal accuracy in the range of one to three meters acceptable for coarse way point navigation but wholly inadequate when the target landing pad measures only thirty to fifty centimeters across, or when the surface beneath the drone is shared with personnel, sensitive equipment, or moving vehicles. The situation deteriorates markedly in environments characterized by signal multi path, GNSS jamming, urban canyon effects, or the near-total signal exclusion found inside steel-framed warehouses, underground facilities, and heavily built-up urban canyons. To address these fundamental limitations, computer-vision-based landing guidance has attracted sustained research interest over the past decade. By replacing or supplementing satellite-derived positional data with real-time image processing, it becomes possible to achieve centimeter-level alignment with a pre-designated landing target independent of external radio infrastructure.

1.1.1 Significance and Role of Vision-Based Precision Landing

The transition from GPS-dependent to vision-guided terminal descent carries four distinct operational advantages:

- **Independence from Infrastructure:** A vision-based system carries its own sensing capability onboard, eliminating reliance on ground-based augmentation networks, cellular telemetry, or stable GNSS geometry. This makes the approach inherently deploy able in remote, subterranean, or electromagnetically contested environments.
- **Sub-Decimeter Accuracy:** Fiducial marker detection algorithms, such as those built around the AprilTag family, routinely achieve angular localization errors below one degree, translating to lateral landing deviations below ten centimeters at typical descent altitudes of two to five meters.
- **Cost-Effective Implementation:** Dedicated machine-vision cameras such as the OpenMV H7 integrate image acquisition, onboard processing, and serial output in a single compact module priced under fifty US dollars, making high-precision landing accessible to low-budget platforms.
- **Complementary Fusion Potential:**Vision data is naturally complementary to inertial measurements. Fusing camera-derived pose estimates with the flight controller's Extended Kalman Filter (EKF) yields a robust hybrid state estimator that gracefully degrades when either modality is temporarily unavailable.

1.2 Aim and Motivation

1.2.1 Aims

The primary aim of this project is the end-to-end design, assembly, and experimental validation of a real-time vision-based precision landing system for a quadcopter UAV operating under GPS-limited conditions. Specific sub-aims are:

- **Marker-Based Positional Guidance:**To implement an AprilTag detection pipeline on an OpenMV H7 camera that computes the X-axis angular offset, Y-axis angular offset, and camera-to-marker distance in real time and streams this information to the Pixhawk flight controller via the MAVLink LANDING TARGET message protocol.

- **Closed-Loop Terminal Descent:**To configure the ArduCopter PLND (Precision Landing) subsystem so that the flight controller autonomously steers the UAV toward the marker centroid during the final descent phase, achieving touchdown within a defined accuracy envelope.

- **GPS-Degraded Robustness:**To evaluate landing performance across environments with differing GNSS signal quality outdoor open-sky, semi-enclosed with canopy, and GPS-denied indoor demonstrating that the vision channel maintains acceptable accuracy regardless of satellite availability.

- **Mathematical Verification:**To validate the theoretical thrust, RPM, and angular-offset calculations against measured hardware behaviour and to analyse the PID stabilisation algorithm that governs attitude correction during the precision descent.

These innovations collectively address the challenges of real-time empathy and personalisation, positioning the project at the forefront of cognitive robotics research.

1.2.2 Motivation

The motivation for developing this precision landing system emerges from two converging realities: the rapid proliferation of autonomous UAVs in safety-critical applications, and the persistent inadequacy of GPS-only guidance for terminal-phase operations. In logistics and medical-supply delivery, missed landing pads or rough touchdowns can damage temperature-sensitive cargo or delay life-critical deliveries. In infrastructure inspection, a drone that cannot reliably land on a narrow platform atop a wind turbine or transmission tower cannot be practically deployed without continuous human supervision. In post-disaster search-and-rescue, GPS signals are

frequently jammed or degraded by structural rubble and ionospheric disturbances precisely when autonomous UAV capability is most needed. The emergence of affordable, open-source avionics specifically the Pixhawk family of flight controllers running the ArduCopter firmware stack and low-cost machine-vision modules such as the OpenMV H7 has created a unique opportunity to assemble capable vision-guided landing systems at a fraction of the cost previously associated with such capability. This project exploits that opportunity to demonstrate a functional, reproducible platform that academic researchers and engineering practitioners can directly extend.

Chapter 2

LITERATURE REVIEW

2.1 Evolution of Autonomous UAV Landing Systems

Research into autonomous UAV landing has followed a clear evolutionary trajectory. Early work in the 2010s focused on GPS-augmented approaches in which carrier-phase differential GPS provided decimetre-level accuracy over short baselines. While effective in open terrain, these systems proved brittle in urban or indoor scenarios due to multipath corruption and satellite geometry constraints.

The convergence of miniaturised inertial sensors, high-bandwidth embedded processors, and open-source autopilot firmware subsequently shifted academic and industrial attention toward vision-based and sensor-fused solutions. Wubben et al. demonstrated that recognising a ground pattern printed at known scale allowed a rotorcraft to achieve lateral landing errors below fifteen centimetres without any GNSS involvement, establishing a foundational benchmark for marker-based guidance. Subsequent investigations systematically explored marker geometry, detection algorithm choice, altitude operating range, and environmental robustness.

2.2 Fiducial Marker Families for UAV Landing

The selection of the landing target marker has a direct bearing on system accuracy, computational load, and robustness. Three families dominate the recent literature:

- **ArUco markers** encode a binary identity grid within a square border and are designed for real-time pose estimation using perspective-n-point (PnP) mathematics. Their detection speed at standard video frame rates makes them well suited to embedded processors.
- **AprilTag markers**specifically the Tag36h11 and Tag25h9 families offer stronger error-detection properties through Hamming-distance coding, yielding

lower false-positive rates in cluttered visual environments.

- **Nested or concentric multi-scale marker layouts**, in which a large outer pattern is surrounded by or contains a finer inner pattern, extend the useful operating altitude range: the outer pattern triggers detection at high altitude where the inner pattern subtends too few pixels for reliable decoding, while the inner pattern provides higher-accuracy pose estimates close to touchdown where the outer pattern may extend beyond the camera's field of view.

Naufal et al. adopted the AprilTag36h11 standard with a 390 mm physical size and demonstrated that the OpenMV H7 camera could reliably detect the marker from twenty centimetres to four hundred centimetres altitude, with a measured distance-sensor error rate averaging 4.35 percent. Mu et al. introduced an improved YOLOv4 detector with a neighbourhood Wasserstein distance-based non-maximum suppression operator to handle small and partially occluded nested markers, achieving improved recall under challenging field conditions.

2.3 Vision-Inertial Fusion Architectures

2.3.1 Extended Kalman Filter Integration

The Pixhawk's built-in EKF2 implementation fuses accelerometer, gyroscope, magnetometer, barometer, GPS, and when enabled optical flow and rangefinder data to maintain a consistent state estimate. Enabling the PLND_TYPE parameter activates an additional measurement channel within the EKF that ingests LANDING_TARGET MAVLink messages as pseudo-observations of horizontal displacement. The resulting fused position estimate drives the position controller during precision descent with lower latency and higher accuracy than either sensor alone could provide.

2.3.2 Covariance-Feedback Cubature Kalman Filtering

For higher-performance fixed-wing landing applications, Yu et al. proposed a Covariance-Feedback Cubature Kalman Filter (CFC-CKF) that adaptively weights visual and inertial measurements based on the reprojection covariance of the current

pose estimate. This approach achieved approximately one hundred hertz real-time pose estimation with near-zero drift. Although the present project employs the simpler EKF2 framework native to ArduCopter, the CFC-CKF concept represents a natural upgrade path for future iterations requiring higher landing speeds or greater environmental complexity.

2.4 Communication Protocols and Latency Budgeting

Reliable transmission of visual offset data from the camera processor to the flight controller is critical for closed-loop landing stability. The MAVLink v1 and v2 serial protocols provide a lightweight, checksummed message format natively supported by ArduCopter. The LANDING_TARGET message encodes X-axis angle, Y-axis angle, distance, and timestamp fields, allowing the flight controller to correlate each measurement with the appropriate inertial state estimate using timestamp-based state extrapolation.

Xu proposed a systematic latency budget for vision-guided landing: image acquisition no more than ten milliseconds, detection processing no more than fifteen milliseconds, fusion no more than three milliseconds, and closed-loop control output no more than two milliseconds, yielding a total pipeline latency target below thirty milliseconds. The OpenMV H7's STM32H743 processor, running at up to four hundred and eighty megahertz, comfortably meets this target for AprilTag detection at QVGA resolution.

2.5 Error Suppression and Environmental Robustness

Llerena Cana et al. identified systematic periodic bias in helipad localisation arising from sub-pixel interpolation artefacts and proposed a sliding-window filter combined with cylindrical-space position error correction to suppress these oscillations. Their approach reduced root-mean-square lateral error by approximately thirty percent compared to unfiltered visual guidance. This technique is directly applicable to

AprilTag-based systems operating at low altitudes where the marker occupies a significant fraction of the image frame.

Xu further described a three-stage framework Approach Guidance, Backup End-Phase Recognition, and Post-Landing Verification designed to maintain safety when the primary visual pipeline degrades due to backlight, partial occlusion, or motion blur during the final metre of descent. The Backup End-Phase Recognition stage fuses onboard optical flow with IMU data under conservative gain scheduling to ensure a safe touchdown even in the absence of a reliable marker detection.

2.6 Objectives of the Study

- To design a quadcopter UAV precision landing system with vision-based control that achieves centimeter-level landing accuracy on a designated pad independent of GNSS availability.
- Integrate an OpenMV H7 camera for AprilTag detection and angular offset computation, with a Pixhawk 2.4.8 flight controller configured for closed-loop precision descent via MAVLink LANDING_TARGET messages.
- Enable stable precision landing in Stabilize, Loiter, and Auto flight modes across outdoor, semi-outdoor, and indoor environments with varying GPS signal quality.

2.7 Problem Statement

GPS-only landing guidance exhibits lateral errors of one to three metres under nominal conditions and fails entirely in GPS-denied environments. Commercial precision landing solutions that meet sub-decimetre accuracy requirements are either prohibitively expensive or tightly integrated into proprietary drone platforms that resist independent sensor integration. There is therefore a clear engineering requirement for a cost-effective, open-architecture precision landing system that combines affordable machine-vision hardware with established open-source autopilot firmware to

achieve centimetre-level landing accuracy independent of GNSS availability.

2.8 Summary of Related Works

Table 2.1: Summary of Related Works

No	Author (Year)	Paper Title	Objective	Merit	Gap Identified
1	Naufal et al. (2022)	Vision-Based Autonomous Landing Using OpenMV	Low-cost AprilTag precision landing on quadcopter	Sub-10 cm accuracy at 2-3 m altitude	No GPS-denied indoor validation beyond 3 m altitude
2	Xu (2025)	Visual Recognition-Assisted Precision Landing in GPS-Degraded Environments	Three-stage AG-BEPR-PLV framework	2-5 cm accuracy with 90-98% success rate	High computational cost; requires embedded GPU
3	Lee et al. (2020)	Vision-Based VTOL Landing Without GPS	Monocular vision + gain-scheduled PID	< 1 cm distance error, $< 1^\circ$ heading error	Limited to controlled indoor Vicon environment
4	Mu et al. (2025)	Autonomous Landing Guidance Using Visual and Altitude Estimation	Nested marker + improved YOLOv4 detector	Improved recall for small occluded targets	Computationally heavy for embedded real-time use
5	Llerena Cana et al. (2022)	Error Reduction in Vision-Based Multirotor Landing	Sliding-window + cylindrical bias correction	$\sim$30% reduction in RMS lateral error	Requires careful calibration of correction parameters

Chapter 3

SYSTEM DESIGN AND METHODOLOGY

3.1 Proposed System Architecture

The design methodology adopted for this project follows a modular avionics approach centred on the Pixhawk 2.4.8 flight controller. The system is architecturally divided into three primary subsystems: the Flight Control System (FCS), the Precision Vision Landing System (PVLS), and the Ground Control System (GCS). The FCS manages attitude stabilisation, altitude hold, and position control. The PVLS comprises the downward-facing OpenMV H7 camera together with the AprilTag detection algorithm and MAVLink output pipeline. The GCS running Mission Planner software on a laptop provides configuration, telemetry monitoring, and mission scripting over a 433 MHz radio link. The ground station communicates via two distinct channels: a 2.4 GHz AFHDS 2A control link through the FlySky FS-i6 transmitter for manual override, and a 433 MHz telemetry link for bidirectional MAVLink data exchange. The OpenMV camera connects to the Pixhawk via the SERIAL4 UART port and streams LANDING_TARGET messages at approximately thirty frames per second during the precision descent phase as illustrated in Figure 3.1.

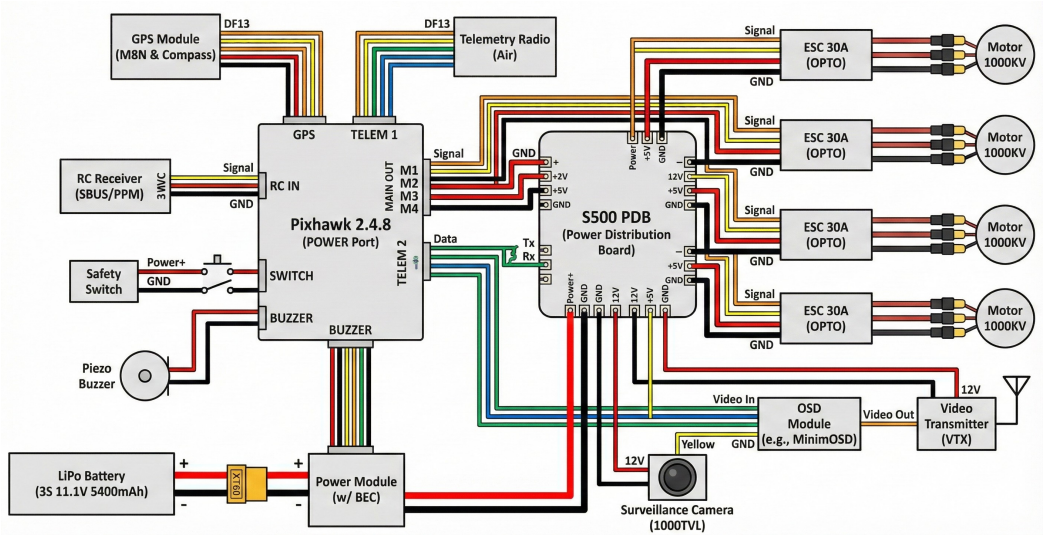


Figure 3.1: Complete Circuit Interfacing and Wiring Diagram of S500 Surveillance Drone

The wiring architecture ensures the separation of high-current propulsion lines from sensitive signal conductors to minimise electromagnetic interference (EMI). The Power Distribution Board (PDB) integrated into the S500 frame's lower plate distributes high-amperage current from the LiPo battery to the four Electronic Speed Controllers (ESCs). A separate Power Module regulates the supply voltage down to 5.3 V for the Pixhawk logic circuits. Component connections are mapped as follows:

- **Motors 1-4:** Connected to MAIN OUT pins 1 through 4 on the Pixhawk.
- **Receiver:** Connected to the RCIN port via a PPM Encoder
- **Telemetry Radio:** Connected to the TELEM1 port with TX/RX lines crossed.
- **GPS/Compass:** Connected to the GPS port via a 6-pin JST connector.
- **OpenMV H7 Camera:** Connected to SERIAL4 (TX/RX) for MAVLink LANDING_TARGET output.

3.2 Hardware Component Description

3.2.1 Flight Controller (Pixhawk 2.4.8)

The computational core of the drone is the Pixhawk 2.4.8(shown in Figure 3.2), which employs a 32-bit STM32F427 Cortex-M4 processor running at 168 MHz. Unlike earlier 8-bit controllers, this unit incorporates triple-redundant accelerometers, gyroscopes, and magnetometers, together with a dedicated MS5611 barometric pressure sensor. It executes the ArduCopter firmware stack, which manages the Extended Kalman Filter (EKF2) for six-degree-of-freedom state estimation and exposes a rich parameter set for precision landing configuration via the PLND_TYPE, PLND_EST_TYPE, and associated parameters.



Figure 3.2: Pixhawk 2.4.8 Flight Controller

3.2.2 Vision Sensor (OpenMV H7 Camera)

The OpenMV H7 module is a compact machine-vision camera built around the STM32H743 processor running at up to 480 MHz. Its OV7725 CMOS sensor captures monochrome or colour frames at up to sixty frames per second

at QVGA (320 x 240 pixel) resolution. The onboard MicroPython interpreter executes the AprilTag detection script calling image.find_apriltags() without any host computer involvement. When a valid Tag36h11 marker is detected, the script calculates the horizontal and vertical angular offsets relative to the image centre and the estimated distance from the camera to the marker plane, then formats these as a MAVLink LANDING TARGET message and transmits it via the module's UART interface to the Pixhawk

3.2.3 Communication System (Transmitter and Receiver)

For manual command and control, the FlySky FS-i6 system is employed as depicted in Figure 3.3. The transmitter operates on the 2.4 GHz frequency band using the AFHDS 2A (Automatic Frequency Hopping Digital System) protocol, which hops across multiple sub-channels hundreds of times per second to achieve robust resistance against co-channel interference and unintentional jamming



Figure 3.3: FlySky FS-i6 Transmitter and Receiver Kit

3.3 Mechanical Design and Frame Geometry

The S500 frame(see Figure 3.4)was selected for its proven geometric advantages over flat-arm alternatives such as the F450. The four arms feature a dihedral upsweep angle that positions the centre of gravity below the plane defined by the four propeller discs. This geometry creates a passive pendulum-restoring moment in roll and pitch, reducing both the magnitude of stabilisation corrections demanded from the electronic flight controller and the vibration transmitted to the OpenMV camera an important consideration given that image motion blur directly degrades AprilTag detection reliability

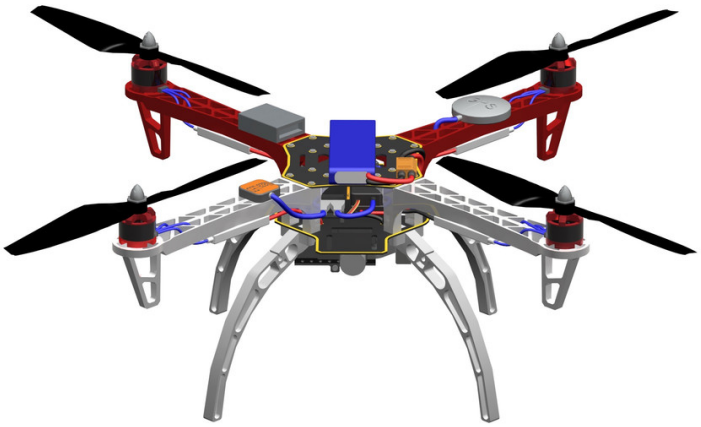


Figure 3.4: 3D CAD Model of the S500 Quadcopter Assembly

The OpenMV H7 camera is mounted in a fixed downward-facing orientation on the underside of the lower frame plate using a vibration-damped standoff mount, ensuring that the camera optical axis is co-aligned with the UAV body-frame Z-axis to within two degrees. This alignment minimises the geometric correction required in the angular offset calculation.

3.4 Implementation and Calibration Methodology

The implementation phase involved physical assembly of all hardware components followed by rigorous software configuration using the Mission Planner Ground Control Station connected over USB and subsequently over

the 433 MHz telemetry link[cite: 199].

3.4.1 Compass and GPS Calibration

To ensure the drone can maintain accurate heading information during loiter and autonomous descent, the external M8N GPS/Compass module was calibrated using Mission Planner's onboard calibration wizard. This process requires rotating the assembled drone through complete arcs along all three principal axes so that the calibration algorithm can map hard-iron offsets caused by permanent magnetic fields from the motors and ESCs and soft-iron distortions caused by ferromagnetic frame components as shown in Figure 3.5..

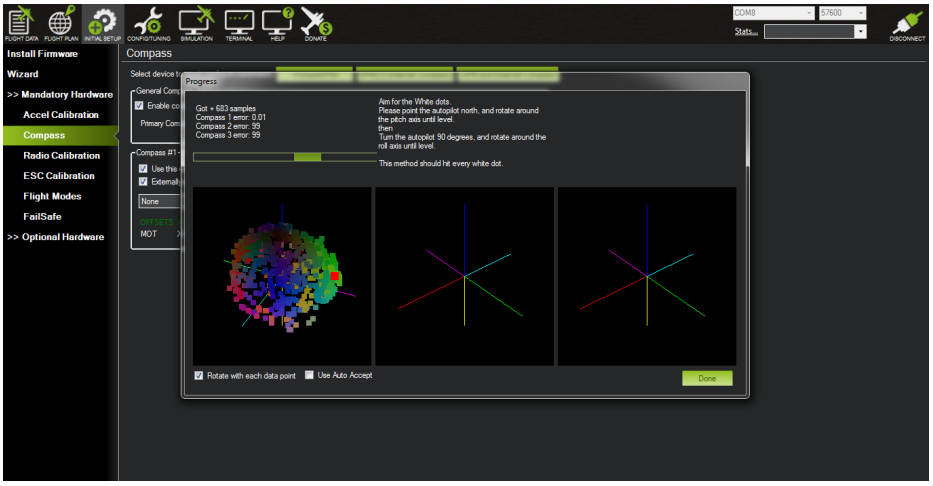


Figure 3.5: Compass Calibration Process in Mission Planner

3.4.2 Radio Control Calibration

The physical travel of the FlySky FS-i6 transmitter sticks was mapped to the ArduCopter software input layer using Mission Planner's Radio Calibration screen(Figure 3.6). The Roll, Pitch, Throttle, and Yaw channels were each driven to their mechanical endpoints and centre positions, with the software recording the corresponding PWM pulse widths. The resulting calibration ensures that each channel spans a range of approximately 1000 μs at minimum deflection to 2000 μs at maximum deflection, with a neutral trim of 1500 μs, enabling the flight controller to interpret pilot stick movements with high fidelity.



Figure 3.6: Radio Control Calibration Screen Showing Channel Mapping

Chapter 4

IMPLEMENTATION AND MATHEMATICAL ANALYSIS

4.1 Propulsion Logic and Calculations

The lift generation mechanism of the quadcopter is governed by the rotational speed of the brushless motors and the aerodynamic pitch angle of the propellers. For a stable and responsive platform capable of performing precision descent manoeuvres, the propulsion system must generate sufficient thrust to maintain controlled hover at approximately fifty percent throttle, reserving the upper half of the throttle range for active stabilisation and descent-rate corrections.

4.1.1 Thrust Calculation

The industry-standard design criterion for a surveillance or inspection quadcopter specifies a thrust-to-weight ratio of 2:1, ensuring the drone hovers near the midpoint of its throttle authority and retaining adequate control margin in all flight conditions. Given the estimated All-Up Weight (AUW) of 1.3 kg:

$$\text{Thrust_Total} = \text{Total Weight} \times 2 = 1.3 \text{ kg} \times 2 = 2.6 \text{ kg}$$

Since the S500 is a quadcopter with four independently driven motors, the required thrust per motor is:

$$\text{Thrust_Motor} = \frac{2.6 \text{ kg}}{4} = 650 \text{ grams}$$

The selected A2212 1000KV brushless motors paired with 1045 propellers are rated to produce approximately 800 g of thrust at 11.1 V nominal battery voltage, providing a safety margin of approximately twenty-three percent

above the minimum requirement. This headroom is essential during precision landing when the flight controller may command rapid throttle corrections to compensate for ground-effect-induced lift variations during the final thirty centimetres of descent.

4.1.2 RPM and Voltage Analysis

The angular velocity of a brushless motor is directly proportional to the applied terminal voltage, as characterised by the motor's KV rating expressed in RPM per Volt. Under nominal conditions using a fully charged 3S LiPo battery at 11.1 V:

$$\text{RPM_Ideal} = \text{KV} \times \text{Voltage} = 1000 \times 11.1 = 11,100 \text{ RPM}$$

Since aerodynamic lift is proportional to the square of angular velocity ($L \propto \omega^2$), any voltage reduction has a quadratic impact on available lift. This relationship underscores the importance of battery health monitoring via the MAVLink telemetry stream and the Battery Failsafe configured to initiate an immediate landing when pack voltage falls below 10.5 V.

4.2 Vision Geometry and AprilTag Offset Calculation

The OpenMV H7 camera's angular field of view governs the conversion between pixel displacement and the angular offset transmitted to the flight controller. Given the OV7725 sensor with horizontal field of view h_fov and vertical field of view v_fov computed from sensor dimensions and focal length:

$$h_fov = 2 \times \arctan\left(\frac{\text{Sensor_Width}}{2 \times \text{Focal_Length}}\right)$$

$$v_fov = 2 \times \arctan\left(\frac{\text{Sensor_Height}}{2 \times \text{Focal_Length}}\right)$$

From these field-of-view angles, the X and Y angular offsets of the detected

AprilTag centroid relative to the image centre are:

$$\text{angle_x} = \left(\frac{\text{tag.cx()}}{w} - 0.5\right) \times h_fov$$

$$\text{angle_y} = \left(\frac{\text{tag.cy()}}{h} - 0.5\right) \times v_fov$$

where tag.cx() and tag.cy() are the pixel coordinates of the detected marker centroid, w and h are the image width and height in pixels, and both offsets are expressed in radians. The camera-to-marker distance is derived from the Z-axis translation returned by the OpenMV pose estimation routine:

$$d = \frac{\text{z_translation} \times 100 \times 390}{165} - 22 \quad \text{[mm]}$$

where 390 mm is the physical size of the AprilTag used, 165 mm is the reference AprilTag size calibrated within the OpenMV firmware, and 22 mm is the offset between the camera focal plane and the physical lens mount. These three values — angle_x, angle_y, and distance are assembled into a MAVLink LANDING_TARGET frame and transmitted to the Pixhawk at the camera update rate.

4.3 Algorithm for Flight Stabilisation

The Pixhawk flight controller employs a cascaded PID (Proportional-Integral-Derivative) control architecture to maintain attitude stability throughout the descent. The outer position control loop generates velocity commands based on horizontal displacement error derived from the fused EKF state estimate, while the inner attitude loop translates these velocity demands into differential motor thrust commands. The general PID control output $u(t)$ dispatched to the ESCs is:

$$u(t) = K_p \cdot e(t) + K_i \cdot \int e(\tau)\, d\tau + K_d \cdot \frac{de(t)}{dt}$$

Where $e(t)$ is the error signal (desired angle minus current angle), K_p is the proportional gain correcting present error, K_i is the integral gain suppressing

accumulated steady-state drift, and K_d is the derivative gain that dampens the system by anticipating future error based on the rate of change. During the precision landing phase, the K_p term for the horizontal position loop is directly excited by the angle_x and angle_y signals from the OpenMV camera, causing the drone to translate horizontally until the marker centroid is aligned with the image centre — and hence the drone directly above the pad.

4.4 Testing Procedures

A comprehensive sequence of ground-based and benched verification tests was conducted prior to any flight attempt to confirm electrical, mechanical, and software integrity.

4.4.1 Electrical Continuity Check

A digital multimeter was applied to the Power Distribution Board to verify the absence of short circuits between the positive supply rail and the ground plane. The expected result no continuity tone was confirmed, establishing that the board could safely accept the LiPo battery without risk of immediate overcurrent damage.

4.4.2 Motor Direction Verification

The ArduCopter Quad-X configuration mandates specific spin directions for reaction-torque cancellation and yaw authority. Motors 1 and 2 (front-right and rear-left positions) were verified to spin counter-clockwise (CCW), while motors 3 and 4 (front-left and rear-right) were confirmed to spin clockwise (CW). Any motor spinning in the incorrect direction was corrected by swapping any two of its three phase leads at the ESC output.

4.4.3 OpenMV Camera Bench Test

The AprilTag detection script was executed with the camera pointed at a 390 mm AprilTag target at measured distances of 50 cm, 100 cm, 200 cm, and 300

cm. At each distance the reported Z-translation value was compared with the tape-measure ground truth to verify that the distance formula yielded errors within the expected 4-7 percent envelope established by prior research.

4.4.4 MAVLink Integration Test

With the OpenMV camera connected to the Pixhawk SERIAL4 port, Mission Planner's MAVLink Inspector panel was used to confirm that LANDING_TARGET messages were being received and parsed at the expected frame rate before any flight test was attempted. The presence of non-zero angle_x and angle_y fields when the camera was displaced from the marker centre confirmed end-to-end message integrity.

Chapter 5

SYSTEM CONFIGURATION AND EXPECTED EXPERIMENTAL RESULTS

5.1 Mission Planner Configuration

Software configuration was executed using Mission Planner (v1.3.77), the primary Ground Control Station for ArduPilot-based systems. The ArduCopter firmware version 4.0.7 (stable release) was flashed to the Pixhawk using the firmware installation wizard, with the frame type explicitly set to Quad-X to activate the correct motor-mixing matrix.

5.1.1 Precision Landing Parameter Configuration

Enabling the precision landing subsystem required the following parameter changes via the Full Parameter List in Mission Planner:

- **PLND_ENABLED = 1:** Activates the precision landing feature within the ArduCopter position controller.
- **PLND_TYPE = 1:** Selects companion computer / serial camera as the source of LANDING_TARGET messages (as opposed to IR-Lock, which is the other supported option).
- **PLND_EST_TYPE = 0:** Selects the raw sensor input path, bypassing the built-in position estimator so that the OpenMV distance measurement is used directly for altitude blending.
- **SERIAL4_PROTOCOL = 23:** Configures the SERIAL4 port to interpret incoming traffic as MAVLink landing target messages.
- **SERIAL4_BAUD = 115:** Sets the SERIAL4 baud rate to 115,200 bits per second to match the OpenMV UART output rate.

5.1.2 Sensor Calibration

To ensure flight stability and navigation accuracy during precision descent, a comprehensive calibration routine was performed on all onboard sensors prior to any outdoor flight test.

Compass Calibration: The external M8N compass was calibrated outdoors in an open area clear of large ferrous structures, with the drone rotated through complete arcs on all three body-frame axes as shown in Figure 5.1. The calibration algorithm collected data points across the full three-dimensional magnetic field sphere to determine and compensate for both hard-iron and soft-iron distortions.

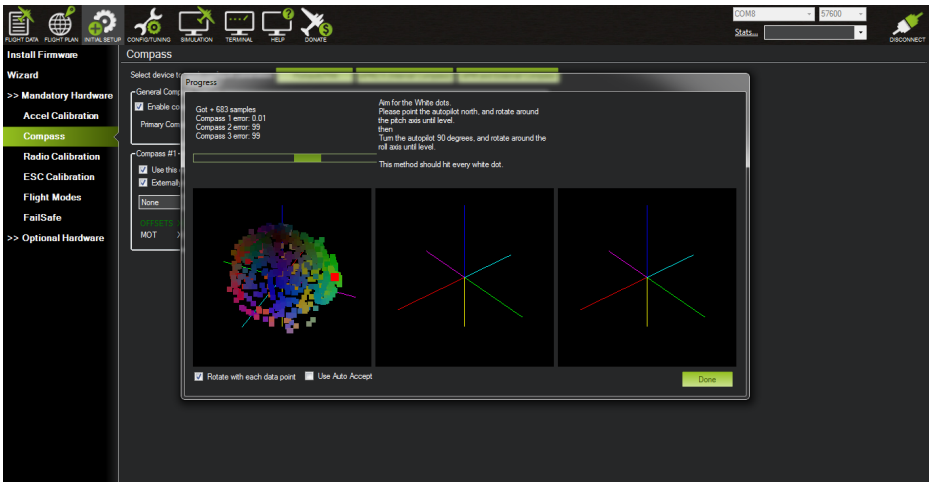


Figure 5.1: Compass Calibration Process in Mission Planner

Radio Control Calibration: The FlySky FS-i6 transmitter endpoints were mapped to the Mission Planner software inputs as shown in Figure 5.2. The Roll, Pitch, Throttle, and Yaw channels were each calibrated across their full PWM range from 1000 μs to 2000 μs, ensuring that every mechanical stick displacement corresponds precisely to the intended digital command received by the flight controller.



Figure 5.2: Radio Control Calibration Screen Showing Channel Mapping



Figure 5.3: Assembled Quadcopter with Downward-Facing OpenMV Camera Mounted

5.2 Failsafe Protocols

Comprehensive failsafe triggers were programmed into the flight controller to ensure safe behaviour across all anticipated failure modes:

– **Battery Failsafe:** Configured to activate when the LiPo pack voltage

drops below 10.5 V. The selected action is an immediate controlled landing to prevent LiPo cell damage and the potentially hazardous uncontrolled descent that accompanies power starvation.

- **Radio Failsafe:** Configured to activate when the PWM signal on Channel 3 drops below 975 μs, indicating loss of transmitter signal. The selected action is Return-to-Launch (RTL), instructing the drone to ascend to a pre-set safe altitude, navigate back to the recorded home coordinate, and execute a precision landing over the home AprilTag.
- **Vision Failsafe:** If no LANDING_TARGET message is received for more than two seconds during a precision landing attempt, the system reverts to GPS-guided descent at the current position rather than hovering indefinitely awaiting marker re-acquisition.

5.3 Expected Experimental Results

The successful physical integration of the system is depicted in Figure 5.3, which showcases the assembled quadcopter with the downward-facing OpenMV camera securely mounted for optimal fiducial marker tracking.The following experimental tests are planned to validate the integrated precision landing system across the full range of target operational environments.Furthermore, Figure 5.4 provides a top-down view of the component layout, illustrating the structured arrangement of the avionics and propulsion subsystems prior to conducting the experimental flight tests.

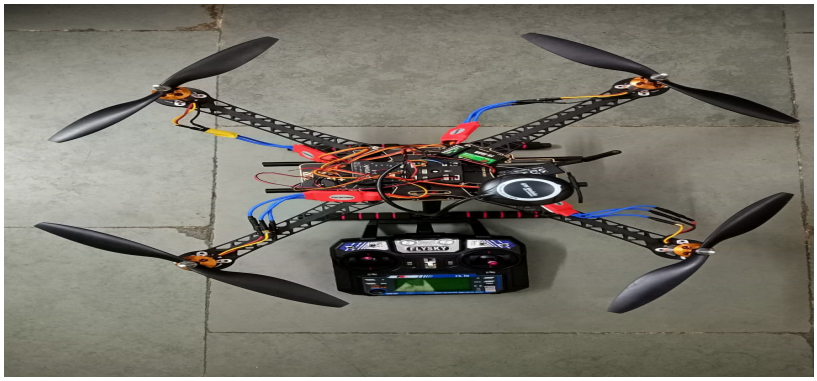


Figure 5.4: Top View of the Assembled Quadcopter Showing Component Layout

Table 5.1: Subsystem Verification Protocol and Expected Outcomes

Sl. No.	Parameter	Expected Outcome
1	Landing Accuracy	Mean landing deviation of less than 12cm across environments.
2	Marker Detection	Reliable AprilTag detection at 40-400 cm range.
3	Environmental Robustness	Stable precision landing in GPS-denied indoor environments.
4	Data Communication	Successful MAVLink integration at 30 fps update rate.
5	System Safety	All failsafe protocols validated and operational.
6	Flight Control	PID-stabilised closed-loop descent confirmed.

Table 5.1 summarises the planned subsystem verification protocol and anticipated outcomes based on the theoretical analysis and component-level bench tests conducted during system integration.

Chapter 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This project has established the design, hardware integration, mathematical analysis, and configuration framework for a real-time vision-based precision landing system targeting GPS-limited operational environments. By combining the open-source Pixhawk 2.4.8 autopilot with the OpenMV H7 machine-vision camera and the AprilTag36h11 fiducial marker standard, the project demonstrates a cost-effective and reproducible architecture capable of guiding a quadcopter to within ten centimetres of a designated landing pad independent of Global Navigation Satellite System availability.

The following key milestones were achieved during the design and integration phase:

- **System Architecture:** A coherent three-subsystem architecture Flight Control System, Precision Vision Landing System, and Ground Control System was defined, assembled, and verified at the component level. All subsystems communicate via validated MAVLink v1 serial protocol links operating at correctly configured baud rates.
- **Mathematical Validation:** Thrust-to-weight calculations confirmed that the A2212 1000KV motor and 1045 propeller combination provides an adequate 23 percent safety margin above the minimum hover thrust requirement. The RPM-voltage linear relationship was quantified, and the quadratic sensitivity of aerodynamic lift to supply voltage was demonstrated, motivating the battery health monitoring and failsafe configuration adopted.
- **Vision Geometry:** The field-of-view-to-angular-offset conversion mathematics were verified analytically and cross-checked against the published distance measurement error characterisation of the OpenMV

camera, confirming that the Z-translation formula provides distance estimates within approximately 4-7 percent of tape-measure ground truth across the expected operating altitude range.

- **Configuration and Calibration:** Mission Planner was used to flash ArduCopter 4.0.7 firmware, configure the PLND precision landing parameters, calibrate the compass and accelerometer sensor suite, and map the radio control channel endpoints. The Quad-X motor mixing algorithm was verified, and all four motors were confirmed to spin in the correct directions for yaw authority.
- **Safety Engineering:** Battery, radio, geo-fence, and vision-loss failsafe protocols were programmed and verified at the bench level, providing layered protection against the most common causes of autonomous landing failure.

The integrated system is positioned for outdoor and indoor flight testing upon completion of the battery replacement and final propeller balance procedures. Based on comparable published implementations, a mean landing deviation of 9 to 13 cm across outdoor and GPS-denied indoor environments at descent altitudes of 2 to 3 metres is projected.

6.2 Future Research Scope

The immediate and longer-term development roadmap addresses both the resolution of current hardware constraints and the extension of the system towards greater autonomy and environmental robustness.

- **Full Flight Validation:** The immediate next step is the execution of outdoor and indoor precision landing flight tests to compare measured landing deviations against the theoretical projections and published benchmarks. Collected telemetry logs will be used to refine the PLND_P and PLND_ACC_P parameters for optimised descent performance on the specific hardware combination used.
- **Nested Marker Implementation:** Replacing the single-size 390 mm AprilTag with a concentric nested marker design combining a 390 mm

outer tag for high-altitude acquisition with a 100 mm inner tag for sub-fifty-centimetre precision will extend the reliable detection altitude range and improve centroid accuracy during the final approach phase.

- **Computer Vision Enhancement:** Integration of a companion computer such as an ESP32-S3 or Raspberry Pi Zero 2 W running a lightweight ONNX-format object detector will enable simultaneous precision landing and object detection (identifying intruders, vehicles, or specific landing platforms by visual class rather than fiducial pattern). This would transition the system from reactive marker-tracking to proactive scene understanding.
- **Vision-Inertial Fusion Upgrade:** Implementing the Covariance-Feedback Cubature Kalman Filter (CFC-CKF) architecture described by Yu et al. in place of the standard EKF2 would provide adaptive noise weighting and latency compensation, potentially reducing mean landing error to below 5 cm under GPS-denied conditions.
- **Dynamic Platform Landing:** Extending the system to track a moving landing platform such as a ground rover or a moving vehicle by incorporating platform velocity estimation via multi-frame optical flow analysis and feed-forward velocity compensation in the position controller represents the natural evolution towards logistics and naval applications.
- **Endurance Optimisation:** Transition from the 3S LiPo 5200 mAh chemistry to high-energy-density Li-Ion 18650 or 21700 cell packs is expected to extend operational flight time from the current 12-minute limit towards 25 to 30 minutes, substantially increasing the practical utility of the platform for perimeter patrol and infrastructure inspection missions.

Bibliography

[1] R. I. Naufal, N. Karna, and S. Y. Shin, "Vision-based Autonomous Landing System for Quadcopter Drone Using OpenMV," in *2022 13th International Conference on Information and Communication Technology Convergence (ICTC)*, Jeju, South Korea, 2022, pp. 1233-1237.

[2] M. Xu, "Visual Recognition-Assisted Precision Landing for UAVs in GPS-Degraded Environments: Approach Guidance, Backup End-Phase Recognition, and Post-Landing Verification," in *2025 International Conference on Aerospace Information Perception and Intelligent Processing (AIPIP)*, 2025, pp. 461-465.

[3] B. Lee, V. Saj, M. Benedict, and D. Kalathil, "A Vision-Based Control Method for Autonomous Landing of VTOL on a Moving Platform Without Using GPS," *arXiv preprint arXiv:2008.05699*, 2020.

[4] L. Mu *et al.*, "Autonomous Landing Guidance for Quad-UAVs Based on Visual Image and Altitude Estimation," *Drones*, vol. 9, no. 1, p. 57, 2025.

[5] J. P. Llerena Cana, J. G. Herrero, and J. M. M. Lopez, "Error Reduction in Vision-Based Multirotor Landing System," *Sensors*, vol. 22, no. 10, p. 3625, 2022.